DSA ki Baithak

CODE BAITHAK

BY NAMAN BHAIYA

Recap (Previous Lecture Connection):

* Structure of a Java Program
* Input & Output in Java
* Variables in Java
* Primitive vs Reference Types
* Identifier Naming Rules
* Keywords in Java

Operators & Operands

Operator → Symbol that performs an operation

Operand → Values on which the operation is performed

Operator

Expression

```
int sum = a + b ;
```

Operands

Expression :

Combination of operators & operands which produces a value

Types of Operators (Based on Operands)

Type	Meaning	Example
Unary	1 operand	a++ , a!
Binary	2 operands	a+b , a/b
Ternary	3 operands	condition ? a : b

Categories of Operators in Java

* Arithmetic Operators (+, -, *, /, %)
* Assignment Operators (=, *=, /=, +=, -=, %=)
* Relational Operators (==, !=, <, <=, >, >=)
* Logical Operators (&&, ||, !)
* Increment / Decrement Operators (++, --)
* Conditional (Ternary) Operator (condition ? :)
* Bitwise Operators (&, |, ^, ~)

will cover later...

Arithmetic Operators (Performs mathematical operations)

Operator	Meaning	Example
+	Addition	a+b
-	Subtraction	a-b
*	Multiplication	a*b
/	Division	a/b
%	Modulus	a%b

Integer / Integer = Integer

Ex: 5/2 = 2

5/2.0 = 2.5

$5/4 = 1$

$5\%4 = 1$

$6\%4 = 2$

$7\%4 = 3$

$8\%4 = 0$

$9\%4 = 1$

$10\%4 = 2$

$11\%4 = 3$

$12\%4 = 0$

$$\begin{array}{r} 1 \\ 4\overline{)5} \\ -4 \\ \hline 1 \end{array}$$

$[0,3]$ $[0, n-1]$

$8\%3 = 2$

$-8\%3 = -2 \rightarrow$ -ve

$= -2 + 3 \rightarrow 1 \rightarrow$ +ve

$8\%-3 = +2 \rightarrow 2$

$-8\%-3 = -2$

$$\begin{array}{r} 2 \\ 3\overline{)8} \\ -6 \\ \hline 2 \end{array}$$

$$\begin{array}{r} 123 \\ 10\overline{)1234} \\ -1230 \\ \hline 4 \end{array}$$

$1234/10 = 123$

$1234\%10 = 4$

$$\begin{array}{r} 12 \\ 100\overline{)1234} \\ -1200 \\ \hline 34 \end{array}$$

$1234/100 = 12$

$1234\%100 = 34$

Modulus ka Gyan...

* Modulus of a number by 'n' always gives answer in the range [0, n-1].

* Modulus of -ve number always gives 2 answers : one +ve & one -ve.

Ex: $-8 \% 3 = -2$
$= 1 \quad (-2 + 3 = 1)$

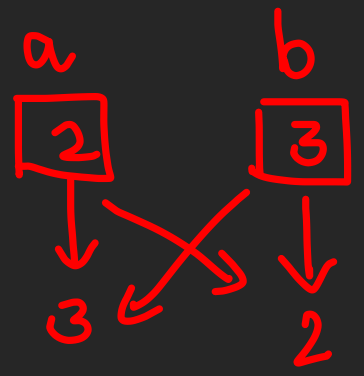
* To get last digit of a number 'n' → $n \% 10$
" " " 2-digits " " " " → $n \% 100$

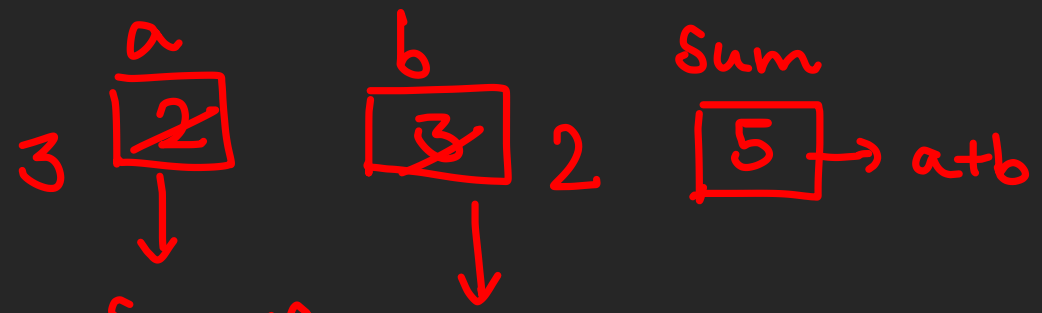
* To remove the last digit of a number 'n' → $n/10$
" " " " 2-digits " " " " → $n/100$

Assignment Operators (Assigns value to a variable)

Operator	Example	Equivalent
=	a=10	assigns value
+=	a += 5	a = a+5
-=	a -= 5	a = a-5
*=	a *= 5	a = a*5
/=	a /= 5	a = a/5
%=	a %= 5	a = a%5

①





sum−a

$(\cancel{a}+b)-\cancel{a}$

b

$5-2=3$

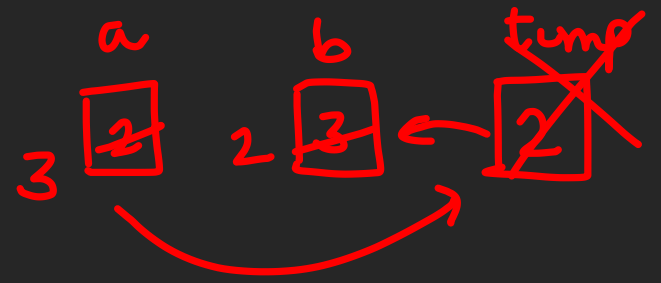
sum−b

$(a+\cancel{b})-\cancel{b}$

a

$5-3=2$

I.



II.

$a+b=sum$

$\boxed{a = sum-b}$

$\boxed{b = sum-a}$

a ← $\underline{b}$ = sum−a

b ← $\underline{a}$ = sum−b

$a = \cancel{2}\ \cancel{5}\ 3$

$b = \cancel{3}\ 2$

$$\boxed{\begin{aligned} a &= a+b \\ b &= a-b \\ a &= a-b \end{aligned}}$$

Relational Operators (Performs the comparison)

Operator	Meaning
>	Greater than
<	Less than
>=	Greater or equal
<=	Less or equal
==	Equal
!=	Not equal

Result → Boolean (true / false)

(2) num → even ✓
×

I. num % 2 = 0 → even ✓
×

II. 5/2 = 2×2 = 4 ✗ → odd

4/2 = 2×2 = 4 ✓ → even

$\boxed{(n/2)*2 == n}$ → even

(3) num → 12345

5

(num/10) % 10 → 1234 5

5

$\boxed{(n/10)\%10}$

(4) $n \rightarrow 123456$

$k = 4, 5$

$\boxed{k=2}$

k^{th}

$\downarrow$

$(n/10)\%10$

$k-1 \Rightarrow x$

$10^2 = 100$

$10^3 = 1000$

$10^{k-1} \rightarrow k-1$

k^{th}

$\%10$

$k = 4$

$pow(10, k-1) = pow(10, 3)$

$\downarrow$

1000

$$\boxed{(n/pow(10, k-1))\%10}$$

$123456/1000 = 123\%10$

$\downarrow$

3

5

$r = 5$

$s = (r = r + 3) + (r += 2)$

$\downarrow$ 8

$\downarrow$ $r = r + 2$

$8 + 2$

10

r	s
~~5~~	18
~~8~~	
10	

$r \rightarrow 10$

$s \rightarrow 18$

Logical Operators (Combines the multiple conditions)

Operator	Meaning
&&	Logical AND
||	Logical OR
!	Logical NOT

Logical AND, Logical OR → Binary operators

Logical NOT → Unary operator

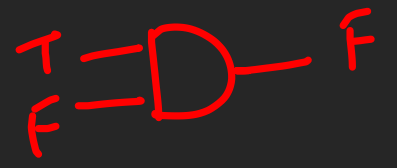
Short Circuit Evaluation occurs in Logical operators.

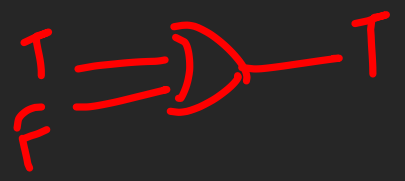
Ex:

false && ____

true || ____

→ skips the second condition





T T > T

F F > F

F → T

Cricket

Ball	Bat	Play
T	F	F
F	T	F
F	F	F
T	T	T

⇓

Logical AND

Halva

Kaju	Badam	Taste
T	T	T
F	T	T
T	F	T
F	F	F

⇓

Logical OR

Increment & Decrement Operators

(Unary Operator)

Increases / Decreases value of a variable by 1

Operator	Example	Equivalent
++	a++ , ++a	a = a + 1
--	a-- , --a	a = a - 1

++5

5 --

X

Pre-Increment/Decrement:

First Modify → then use (++a / --a)

++a

a=a+1

Post-Increment/Decrement:

First use → then Modify (a++ / a--)

~~++5~~

5 = 5+1

6

$x = 3$

$y = x{+}{+} + {+}{+}x + {-}{-}x + x{-}{-}$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$3 \quad 5 \quad 4 \quad 4$

x	y
~~3~~	16
~~4~~	
~~5~~	
~~4~~	
3	

$x \rightarrow 3$

$y \rightarrow 16$

(7) $exp = 5 + 9 * 10/3 - 2$ $\longrightarrow$ BODMAS

$\cancel{= 5 + 9 * 3 - 2}$

$\cancel{= 5 + 27 - 2}$

$\cancel{= 32 - 2 = 30}$

$= 5 + 90/3 - 2$

$= 5 + 30 - 2$

$= 35 - 2$

$= 33$

BODMAS vs Programming

Maths → BODMAS

Programming → Operator Precedence, Associativity

Operators Precedence

Higher Precedence → executed first

	Operators
(Highest)	++, --, !
	*, /, %
	+, -
	<, >, <=, >=
	==, !=
	&&
	||
(Lowest)	=, +=, -=, *=, /=

Associativity

(Decides the evaluation order of same precedence operators)

Operator Type	Associativity
Arithmetic	L to R
Relational	L to R
Logical	L to R
Assignment	R to L
Unary	R to L

L → Left

R → Right

Longest Valid Token Rule

(8) $p = 5$

$q = p+++p++p$

$p++ \rightarrow 5$, $+\ p++ \rightarrow 6$, $+\ p \rightarrow 7$

p	q
~~5~~	18
~~6~~	
7	

$p \rightarrow 7$

$q \rightarrow 18$

(9) $val = 2$

$ans = ++val++$

$\rightarrow$ ~~++val++~~

$\rightarrow$ $++(val++)$ $\rightarrow$ ~~++2~~ $\rightarrow$ 3

val
~~2~~
3

$\rightarrow$ Error